

ZS-T10

The $T_Y \rightarrow X$ Sector-Conversion Operator:

A Geometry-First ANC Correction $\delta \ln \gamma_b = -\kappa^2$ for ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$, a Per-Nucleus Falsification Programme, the ${}^8\text{B}$ Consistency Test, and the Closed $C_0 \leftrightarrow \kappa^2$ RG-Running Matching

Kenny Kang

Z-Spin Cosmology Collaboration

Theme/Code: Translational [ZS-T] | Paper 10 | ZS-T10 v2.1 | March 2026

Verification: 35 checks PASS | Gate Li-2 FLAGGED (nonlinear ${}^7\text{Li}$ tension) | Gate S-T10.1 CLOSED-CONDITIONAL (§10) | Zero Free Parameters

$A = \delta_X \cdot \delta_Y = (5/19)(7/23) = 35/437$ (polyhedral, ZS-F2, BBN-independent); $Q = 11$; $\kappa^2 = A/Q$

Epistemic Status Legend

Status	Definition
PROVEN	Mathematical theorem; standard math alone, machine-verifiable.
DERIVED-CONDITIONAL	DERIVED conditional on a listed axiom set or upstream theorem.
DERIVED-PERTURBATIVE	Derived to all orders in perturbation theory; non-perturbative completion deferred (weak-curvature scope). [Added v2.1, inherited from ZS-M13/M6 §7A.]
HYPOTHESIS-strong	Structurally motivated conjecture with corpus-PROVEN quantum; forward derivation OPEN.
IMPORTED	Result proved externally and used here; full citation given.
TESTABLE	Pre-registered prediction with explicit falsification protocol.
FLAGGED / TENSION	Verification check that did not pass cleanly; honestly registered.
NON-CLAIM	Explicit declaration of what is NOT asserted; bounds the framework’s scope.
OPEN	Recognized gap honestly registered; resolution path given where available.
CLOSED-CONDITIONAL	Formerly OPEN gate now resolved at DERIVED-CONDITIONAL level; residual narrow caveat registered. [Added v2.1.]

§0. Abstract

We present the $T_Y \rightarrow X$ sector-conversion operator and its single, geometry-fixed nuclear prediction. The governing constant $A = \delta_X \cdot \delta_Y = (5/19)(7/23) = 35/437$ is derived from polyhedral curvature asymmetry (ZS-F2) and was fixed by the H_0 tension ($H_{0\text{local}}/H_{0\text{CMB}} = e^A$) and the baryon-to-photon ratio η_B BEFORE any nuclear input — it is not tuned to lithium (§1.1).

Standard-physics statement: the ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ radiative-capture rate receives a geometric correction $\delta \ln \gamma_b = -\kappa^2 = -A/Q \approx -0.73\%$ on the ${}^7\text{Be}$ asymptotic normalization coefficient (ANC), equivalent to a fixed shift of the ${}^3\text{He}$ – ${}^4\text{He}$ cluster-EFT short-range Wilson coefficient C_0 (§2.2).

Z-Spin prediction (now DERIVED-CONDITIONAL, v2.1): $S_{34}(E)$ in the BBN Gamow window (0.1–0.5 MeV) is systematically lower by $\approx 0.7\text{--}1.5\%$, at the precision frontier of LUNA-MV (§4.2). In

v2.0 this was an interesting falsifiable hypothesis; v2.1 closes the $C_0 \leftrightarrow \kappa^2$ matching, promoting it to a parameter-free DERIVED-CONDITIONAL number that remains TESTABLE at LUNA-MV-class systematics ($\approx 1\text{--}2\%$).

Immediate consistency test: a universal binding shift $-\dim(Z)\kappa^2$ would unbind ${}^8\text{B}$ ($\delta Q/Q = -401\%$) and destabilize ${}^9\text{C}$ (-44%); since both are observed (AME2020 [12]), the action must be bottleneck-selective (NC-T10.5). The master-equation slow eigenvalue is $\lambda_{\text{slow}} = -2\mathbf{A}/\mathbf{Q} = -\dim(Z)\kappa^2$ (§2.1). The structural anti-numerology Bayes factor is $\ln\Lambda = 7.2$ (very strong, not decisive; §6.5).

v2.1 closure: the $C_0 \leftrightarrow \kappa^2$ matching is closed at DERIVED-CONDITIONAL via Theorem T10.6 (κ^2 Non-Running, §10.2), which establishes the RG-invariance of $\kappa^2 = \mathbf{A}/\mathbf{Q}$ from M_p to the nuclear scale via the corpus Continuum Perturbative Protection Theorem (ZS-M13/M6 §7A, PROVEN-PERTURBATIVE) and the Appelquist–Carazzone decoupling theorem [15]. The closure changes NO numerical value (downstream-inert; §10.6). The PRyMordial gate remains the principal caution: the physical ${}^7\text{Li}$ closure is coordinate-only (linear $+0.3\sigma$ vs nonlinear -3.3σ , Gate Li-2 FLAGGED) — v2.1 does NOT claim the lithium problem is solved. Zero free parameters.

§1. Introduction

The cosmological lithium problem — the factor-3 excess of standard-BBN ${}^7\text{Li}/\text{H}$ over the Spite-plateau value — has resisted nuclear, astrophysical, and beyond-Standard-Model solutions for three decades. This paper does not claim to solve it. Instead it isolates a single, geometry-fixed correction to one reaction, states it in standard nuclear-physics language, and gives the one measurable number by which it can be falsified. ZS-T9 left two items OPEN: a forward operator carrying the conversion eigenvalue onto the ${}^7\text{Be}$ matrix element, and the O-F19.6 absolute scale. ZS-T10 establishes the operator and executes a PRyMordial stress-test that we report honestly, tension included.

§1.1. Prior Derivation of $\mathbf{A} = 35/437$ and Its BBN Independence

$\mathbf{A}$ is not a lithium fit. It is the product structure $\mathbf{A} = \delta_X \cdot \delta_Y = (5/19)(7/23) = 35/437$, computed from the face–vertex curvature asymmetry of the truncated-octahedron $\times$ truncated-icosahedron register geometry (ZS-F2, LOCKED), with no nuclear input. Its chronological priority is explicit:

Table 1.1. Chronological priority of $\mathbf{A}$ (geometry and cosmology fix $\mathbf{A}$ before any nuclear use).

Stage	Constraint that fixed $\mathbf{A}$	Source / External reference
1. Geometry	$\mathbf{A} = \delta_X \cdot \delta_Y$ from polyhedral curvature asymmetry	ZS-F2 (LOCKED)
2. Cosmology	$H_0/H_{\text{CMB}} = e^\Lambda$ (Hubble tension, 7.4%); $e^\Lambda = 1.0834$, pull 0.06σ	ZS-F3, ZS-U4; [13,14] (external)
3. Baryogenesis	η_B via measure-projection weight; Planck 2018 [14] $+0.07\sigma$	ZS-F2 §9–10; [14] (external)
4. THEN BBN	Applied to ${}^7\text{Be}$ here, without adjustment	This work

An external reader can verify that $\mathbf{A}$ was fixed by CMB/ H_0 and η_B data upstream of the present nuclear application: the lithium use is a prediction, not post-hoc tuning. The two external references [13,14] allow independent confirmation that e^Λ matches the observed H_0 ratio.

§1.2. Standard-Physics Translation (no Z-Spin vocabulary)

For readers outside the Z-Spin programme, the claim of this paper is: the ${}^3\text{He}\text{--}{}^4\text{He} \rightarrow {}^7\text{Be}$ radiative-capture amplitude receives a geometric correction $\delta \ln \gamma_s = -\mathbf{A}/\mathbf{Q} \approx -0.73\%$ on the asymptotic

normalization coefficient, arising from a Planck-scale sector-mediation constraint that is geometrically fixed (zero free parameters). In conventional terms this is a fractional shift of the ${}^7\text{Be}$ ground-state ANC, equivalently a shift of the leading short-range (contact) Wilson coefficient C_0 in halo/cluster EFT. It predicts a small, definite reduction of $S_{34}(E)$ at BBN energies and, under the linear sensitivity of Dent et al. [1], a downward ${}^7\text{Li}$ shift toward the Spite plateau. Whether that shift is the full lithium resolution is left open; the measurable content is the S_{34} reduction itself, now derived parameter-free in §10.

§2. The $T_{Y \rightarrow X}$ Operator

The ZS-Q7 inter-sector generator ($W_{AB} = \dim(B) \cdot A/Q$, the Fermi-golden-rule rate into sector B) over the coarse-grained sectors (X,Z,Y) factorizes as

$$\lambda(Q\lambda + \dim(Z)A)(Q\lambda + (\dim X + \dim Y + \dim Z)A)/Q^2 = 0,$$

giving $\lambda_{\text{slow}} = -\dim(Z)A/Q$ (the negative Z-bottleneck entry-rate, structural for arbitrary dimensions) and $\lambda_{\text{fast}} = -A$ (because $\dim X + Y + Z = Q$). Hence

$$T_{Y \rightarrow X} = -(\dim(Z) \cdot A/Q) \cdot P_{\text{slow}}, \quad \text{action level } \lambda_{\text{slow}} = -2A/Q = -70/4807,$$

lifted to DERIVED-CONDITIONAL by the Gorban–Radulescu–Zinovyev limiting-step theorem [3] (a multiscale linear network’s slow relaxation reduces to a single limiting step).

§2.1. The $\lambda_{\text{slow}} = -\dim(Z) \cdot \kappa^2$ Identity (DERIVED-CONDITIONAL)

With $\kappa^2 = A/Q$ (Register-Total Normalization, ZS-M6 §2.2 PROVEN): $\lambda_{\text{slow}} = -2A/Q = -2\kappa^2 = -\dim(Z) \cdot \kappa^2$ (exact rational $-70/4807$). Numerically identical to v1.0 — zero downstream change — but expressing the conversion eigenvalue through the PROVEN cross-sector coupling κ .

§2.2. What $T_{\{Y \rightarrow X\}}$ Is in Conventional Nuclear Language

In ${}^3\text{He}$ – ${}^4\text{He}$ cluster EFT the ${}^7\text{Be}$ ground state is a shallow two-body bound state with binding momentum $\gamma_b = \sqrt{2\mu B}$ and ANC C . The operator acts as $\gamma_b \rightarrow \gamma_b(1 - \kappa^2)$, i.e., a fractional reduction of the binding momentum by $\kappa^2 = 0.728\%$. Because the external direct-capture cross-section scales with the ANC, this is equivalent, in standard EFT bookkeeping, to a shift of the leading short-range contact-term Wilson coefficient C_0 that reproduces the same $\delta C/C$. No new operator structure is introduced; the Z-Spin content is the fixed numerical value $\kappa^2 = A/Q$ of that shift.

§2.3. Why the Action Is Bottleneck-Selective (theoretical basis, HYPOTHESIS-strong)

The ${}^8\text{B}/{}^9\text{C}$ test (§4.1) shows empirically that the operator cannot act as a universal binding rescaling. The theoretical reason follows from the same Gorban–Radulescu–Zinovyev limiting-step theorem [3] that lifts §2: in a multiscale linear network with well-separated rates, the reduction collapses the dynamics onto a SINGLE limiting (slowest) step. The action level of $T_{\{Y \rightarrow X\}}$ is precisely λ_{slow} , i.e., that limiting step.

In the BBN mass-7 subnetwork the limiting step is ${}^7\text{Be}$ production and destruction (${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$; ${}^7\text{Be}(n, p){}^7\text{Li}$); ${}^8\text{B}$ enters only through the side branch ${}^7\text{Be}(p, \gamma){}^8\text{B}$, which is quantitatively negligible: at BBN temperatures ($T \sim 0.1$ – 1.0 MeV) the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction rate is $\sigma v \sim 10^{-27} \text{ cm}^3 \text{ s}^{-1}$, roughly 10^5 – 10^6 times smaller than the ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ rate at the same epoch [9]. The operator therefore acts on the ${}^7\text{Be}$ channel and not on ${}^8\text{B}$ ’s binding. This identification of “the limiting step” with ${}^7\text{Be}$ specifically is

HYPOTHESIS-strong; in v2.1 its forward derivation is materially advanced by the closure of §10 (the rate-limiting bottleneck inherits the protected, RG-invariant κ^2 shift).

§3. The ${}^7\text{Be}$ Cluster-EFT Scaffold and the $\delta\ln\gamma_s = -\kappa^2$ Foothold

${}^3\text{He}-{}^4\text{He}$: $\mu = 1601.6$ MeV, $B = Q_{\text{sep}} = 1.586$ MeV (AME2020 [12]), $\gamma_b = \sqrt{(2\mu B)/\hbar c} = 0.361$ fm $^{-1}$ ($\equiv 71.28$ MeV). Since $\gamma_b \propto \sqrt{B}$, the operator gives $\delta\ln\gamma_b = \frac{1}{2}\delta\ln B = \frac{1}{2}\lambda_{\text{slow}} = -A/Q = -\kappa^2$ [v2.0: HYPOTHESIS-strong; v2.1: DERIVED-CONDITIONAL via §10]. The effective range parameter $\gamma_b r_0 \approx 0.40-0.50$ is consistent with the ${}^3\text{He}-{}^4\text{He}$ scattering data compiled in Zhang, Nollett, Phillips (2020) [4]. The ANC modulation rate of ${}^7\text{Be}$ equals the PROVEN cross-sector coupling κ^2 ; the Level-3 chain ZS-S10 action $\rightarrow \kappa$ gauge bridge $\rightarrow \kappa^2$ ANC modulation $\rightarrow \delta\ln B_{7\text{Be}} = -2\kappa^2$ (E1 phase space) is now completed in §10. The E1 capture carries $E_\gamma^3 \propto Q^3$; with the PRyMordial forward slope +0.969 and amplification $B_{\text{tot}}/Q = 23.7$ the Dent sensitivity reconstructs to +69 (cf. Dent +81 [1]).

§4. PRyMordial Stress-Test

Standard-BBN baseline (PRyMordial small network, NACRE-II): ${}^7\text{Li}/\text{H} = 5.49 \times 10^{-10}$. $B_{7\text{Be}}$ enters via the ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ detailed-balance Q-value and forward S-factor ($\partial\ln{}^7\text{Li}/\partial\ln$ rate = +0.969).

Table 4.1. PRyMordial ${}^7\text{Li}$ response (linear vs nonlinear injection).

Method	Response	${}^7\text{Li}$ outcome
LINEAR Dent +81	$\Delta\ln{}^7\text{Li} = -1.18$	1.69×10^{-10} (+0.3 σ)
NONLINEAR (Q^3)	eff. sens. +150	0.62×10^{-10} (−3.3 σ)
D/H, $Y_{\square}$ (all runs)	$\Delta < 0.07\%$	decoupled ✓

The linear/nonlinear disagreement is registered as Gate Li-2 (FLAGGED): the physical ${}^7\text{Li}$ closure is coordinate-only (NC-T10.1). v2.1 does not change this: the §10 closure addresses the ANC matching (Gate S-T10.1), not the ${}^7\text{Li}$ closure (Gate Li-2).

§4.1. Per-Nucleus Falsification Table and the ${}^8\text{B}/{}^9\text{C}$ Consistency Test (P-T10.1, TESTABLE)

If $\delta\ln B_{\text{tot}} = -2A/Q$, the implied Q-value shift is $\delta Q/Q = -2A/Q \cdot (B_{\text{tot}}/Q)$, with nucleus-specific amplification:

Table 4.2. Per-nucleus consistency test of a (hypothetical) universal binding shift.

Nucleus	Cluster channel	B_{tot}/Q	$\delta Q/Q$ @ $-2A/Q$
${}^7\text{Be}$	${}^3\text{He}+{}^4\text{He}$	23.7	−34.5%
${}^6\text{Li}$	$d+{}^4\text{He}$	21.7	−31.6%
${}^9\text{Be}$	${}^8\text{Be}+n$	34.9	−50.9%
${}^9\text{C}$	${}^8\text{B}+p$	30.1	−44.0%
${}^8\text{B}$	${}^7\text{Be}+p$	275	−401% (unbinds)

Immediate consistency test (no new experiment needed). ${}^8\text{B}$ is bound by only $S_p = 0.137$ MeV (AME2020 [12]), so a universal $-2A/Q$ binding shift implies $\delta Q/Q = -401\%$, which would completely unbind ${}^8\text{B}$; the same shift destabilizes ${}^9\text{C}$ by −44%. Both nuclei are observed to exist. Therefore the operator CANNOT act as a universal binding rescaling — a literal universal action is immediately falsified by existing nuclear data — and must act only on the rate-limiting (bottleneck) channel, exactly

the limiting-step selection of §2 (NC-T10.5). The ^8B datum thus serves simultaneously as a falsification of the naïve reading and as positive evidence for the bottleneck-selective structure.

§4.2. The Z-Spin Prediction and Its Experimental Programme (DERIVED-CONDITIONAL / TESTABLE)

The bottleneck-selective reading applied to ^7Be gives the one clean, standard-physics prediction of this paper: $\delta \ln \gamma_b = -\kappa^2 = -0.728\%$ on the ^7Be ANC. For the external (peripheral) direct-capture reaction the low-energy S-factor is proportional to the ANC squared, $S_{34}(E) \propto C^2$ (ANC method [6,7]; halo-EFT leading order [5]), so $\delta \ln S_{34} = \delta \ln C^2$. In halo EFT the bound-state normalization gives $C^2 \propto \gamma_b/(1 - \gamma_b r_0)$ (Spärgberg–Capel–Baye [17]), hence $d \ln C^2/d \ln \gamma_b = 1/(1 - \gamma_b r_0)$, which runs from 1 at leading (zero-range) order to ≈ 2 with the ^3He – ^4He effective range ($\gamma_b r_0 \approx 0.40$ – 0.50 , consistent with Zhang et al. 2020 [4]). Thus

$$\delta \ln S_{34}(E) \approx -0.7\% \text{ (LO) to } -1.5\% \text{ (with effective range), } E \text{ in the BBN Gamow window (0.1–0.5 MeV).}$$

In v2.0 this was an interesting falsifiable hypothesis. In v2.1, §10 closes the $C_0 \leftrightarrow \kappa^2$ matching: every input ($\kappa^2 = A/Q$ geometric; γ_b from AME2020; $\gamma_b r_0$ from Zhang et al.) is LOCKED or external-measured, so the prediction is now parameter-free DERIVED-CONDITIONAL. It still lies at or below current $\approx 4\%$ precision (LUNA activation at 127–169 keV [7]) but within reach of the LUNA-MV 3.5 MV underground accelerator now re-measuring $^3\text{He}(\alpha, \gamma)^7\text{Be}$. Honest magnitude assessment: confirmation requires LUNA-MV-class systematics at ≈ 1 – 2% ; we do NOT claim the shift is already measured (NC-T10.7).

Table 4.3. Experimental programme for the Z-Spin nuclear prediction.

Prediction	Test	Facility	Window
S_{34} low by $\sim 1\%$ ($\delta \ln \gamma_b = -\kappa^2$)	$^3\text{He}(\alpha, \gamma)^7\text{Be}$ S-factor, 0.1–0.5 MeV	LUNA-MV (Gran Sasso)	running ≥ 2019
^6Li ANC / S-factor shift	$^2\text{H}(\alpha, \gamma)^6\text{Li}$ direct capture	LUNA (Gran Sasso)	measured/ongoing
^8B unbinds under universal action	AME2020 $^8\text{B}/^8\text{C}$ exist $\rightarrow$ selective action	AME2020 [12]	immediate ✓

§5. The O-F19.6 Absolute-Scale Closure

Connection to the main result. O-F19.6 closes the second ZS-T9 OPEN item and is included because it fixes the absolute information scale of the SAME Z-bottleneck through which $T_{\{Y \rightarrow X\}}$ operates: both descend from the register equilibrium $p_{\text{eq}} = (3, 2, 6)/11$. The $\frac{1}{2} \ln 2$ entropy per $X \rightarrow Y$ conversion is the information-theoretic counterpart of the operator’s fixed $-2A/Q$ action level — it is what licenses treating that level as absolute rather than merely relative: the same geometric equilibrium that determines λ_{slow} also determines the per-transition entropy quantum $\frac{1}{2} \ln 2$, providing mutual consistency of the two results.

With $p_{\text{eq}} = (3, 2, 6)/11$ (L12 PROVEN) the modular Hamiltonian $K = -\ln p_{\text{eq}}$ has $X \rightarrow Y$ gap $\Delta K = -\ln 2$; the half-gap $\frac{1}{2} \ln 2 = \psi_{\text{KMS}}$ reproduces $\tanh(2\psi_{\text{KMS}}) = 3/5$ (Theorem F19.6). The Type II $_{\infty}$ trace weight being fixed by p_{eq} , $\Delta S_{\text{gen}} = \frac{1}{2} \ln 2 = 0.5$ bit is ABSOLUTE, with register constant $\ln Q = \ln 11$. DERIVED-CONDITIONAL on the $(3, 2, 6) \leftrightarrow$ degenerate-clock identification [2].

§6. Anti-Numerology Monte Carlo

Part A (500k): $\lambda_{\text{slow}} = -\dim(Z)A/Q$ in 100.0000% of random configs (max err 1.8×10^{-15}). Structural identity, not fit.

Part C (500k): Arbitrary-rate chains reproduce $-2A/Q$ only 0.54% — the Fermi-golden-rule rate law is required.

Part B (2M + 414 quanta): Value-fit selectivity 3.9% (modest); at fixed $-2A/Q$ only $B_{7\text{Be}}$ lands (others +6 to +20 σ).

§6.4. Structural vs Statistical Anti-Numerology

$-2A/Q$ is a fixed eigenvalue, not a fitted parameter, so the statistical fit-test (Part B, 3.9%) is the wrong instrument. The operative evidence is structural: Part A (100% structural identity) and Part C (0.54% rate-law necessity). A deterministic eigenvalue carries no probability factor; forming a “combined probability” by multiplying in a zero is a numerology-adjacent error and is explicitly NOT done here.

§6.5. Bayes Factor for the Structural Evidence (DERIVED-CONDITIONAL)

Explicit hypotheses. H_0 (null): the action level $-2A/Q$ landing on the ${}^7\text{Be}$ channel is a chance coincidence of an arbitrary inter-sector rate structure. H_1 (Z-Spin): the structural derivation predicts $-2A/Q$ (the master slow eigenvalue) acting on the mass-7 bottleneck. The Bayes factor is $\Lambda = P(\text{data}|H_1)/P(\text{data}|H_0)$. Under H_1 the prediction is deterministic, $P(\text{data}|H_1) \approx 1$. Under H_0 the data require two independent coincidences: an arbitrary rate structure reproducing $-2A/Q$ (MC Part C, $p_C = 0.0054$) AND that structure selecting the ${}^7\text{Be}$ channel among the seven decoupled binding channels ($p_{\text{chan}} = 1/7$). Assuming independence: $P(\text{data}|H_0) = p_C \cdot p_{\text{chan}} \approx 7.7 \times 10^{-4}$, giving $\Lambda \approx 1296$, $\ln \Lambda = 7.2$.

Honest caveats. This is an order-of-magnitude likelihood ratio, not a full Bayes factor: (i) the independence of value- and channel-coincidences is assumed; (ii) $P(\text{data}|H_1) = 1$ ignores the prior plausibility of H_1 itself; (iii) p_C and p_{chan} are Monte-Carlo point estimates. On the Jeffreys scale $\ln \Lambda = 7.2$ is “very strong” ($\ln \Lambda > 5$) but NOT “decisive” ($\ln \Lambda > 10$). Status: DERIVED-CONDITIONAL on those assumptions.

§7. Falsification Gates

Gate	Condition	Status
Li-1	PRyM reproduces Dent +81 at linear order	PASS (+69 LO)
Li-2	Nonlinear $B_{7\text{Be}}$ injection matches linear closure	FLAG / TENSION
Li-3	D/H and $Y_{\square}$ decoupled ($\Delta < 0.07\%$)	PASS
P-T10.1	${}^8\text{B}$ unbinds / ${}^9\text{C}$ destabilized under universal action — FALSIFIED by AME2020	PASS-consistency (selective)
S34-1	$S_{34}(0.1\text{--}0.5 \text{ MeV})$ low by $\sim 1\%$ ($\delta \ln \gamma_{\square} = -\kappa^2$); parameter-free after §10	TESTABLE (LUNA-MV)
S-T10.1	Cluster-EFT $C_0 \leftrightarrow \kappa^2$ matching (explicit RG running from Planck to nuclear scale)	CLOSED-CONDITIONAL (§10, v2.1)

Gate S-T10.1 status change (v2.1): CLOSED-CONDITIONAL. The matching is executed in §10 at DERIVED-CONDITIONAL level. A residual narrow OPEN remains — the non-perturbative completion of the protection theorem (Theorem T10.6(i) is PROVEN-PERTURBATIVE only) — registered as Gate S-T10.2 below. This residual does not affect the nuclear-scale numerical content (the nuclear scale is deep in the weak-curvature regime) and does not block downstream T-papers.

Gate	Condition	Status
S-T10.2	Non-perturbative (strong-curvature) protection of κ^2 against running; needed for unconditional DERIVED	OPEN (narrow; non-blocking)

§8. Non-Claims

NC-T10.1. The ${}^7\text{Li}$ binding-channel closure is coordinate-only; Gate Li-2 does not validate it as a physical mechanism. The §10 closure does NOT change this: it addresses the ANC matching, not the ${}^7\text{Li}$ closure.

NC-T10.4. No new free parameter; $\mathbf{A} = \delta_x \delta_y$, $\mathbf{Q}$, $\dim Z$, $\kappa^2 = \mathbf{A}/\mathbf{Q}$ all LOCKED. The §10 closure introduces no parameter (NC-T10.9).

NC-T10.5. The operator acts bottleneck-selectively, NOT as a universal binding rescaling (${}^8\text{B}/{}^9\text{C}$ test forbids the latter).

NC-T10.6. The cross-corpus invariant is $\dim(Z) = 2$, NOT the rational $-2\mathbf{A}/\mathbf{Q}$.

NC-T10.7. The predicted S_{34} shift ($\sim 1\%$) is not claimed to be already measured; it is a target for LUNA-MV-class precision.

NC-T10.8 (v2.0, superseded by §10). v2.1 supersedes this non-claim: the matching is now claimed and executed in §10 at DERIVED-CONDITIONAL level. The original v2.0 wording is preserved in Appendix B (no-deletion rule) for the audit record.

NC-T10.9 (v2.1). The §10 closure does NOT introduce any new free parameter, field, or postulate. It is a matching/non-running statement built entirely on LOCKED constants ($\mathbf{A}$, $\mathbf{Q}$, κ^2), corpus-PROVEN theorems (ZS-M13/M6 §7A, ZS-Q7, ZS-S10), and externally PROVEN results [15,16,17]. It does NOT claim a full first-principles loop computation of the ${}^7\text{Be}$ ANC from M_p ; it claims RG-invariance of the geometric ratio κ^2 and the consequent scale-stability of the physical observable.

NC-T10.10 (v2.1). The closure does NOT claim the lithium problem is solved, nor does it lift Gate Li-2. The Z-Spin nuclear content is the S_{34} reduction, not the full ${}^7\text{Li}$ resolution.

§9. Conclusion, $\dim(Z)=2$ Convergence

ZS-T10 isolates one geometry-fixed, standard-language prediction — $\delta \ln \gamma_b = -\kappa^2$ on the ${}^7\text{Be}$ ANC, hence S_{34} low by $\sim 1\%$ — with $\mathbf{A}$ fixed by CMB/ H_0 [13,14] long before any nuclear use, and an immediate ${}^8\text{B}/{}^9\text{C}$ consistency test. The cross-corpus signal is the cardinal-2 invariant $\dim(Z) = 2$, not the rational $-2\mathbf{A}/\mathbf{Q}$ (NC-T10.6):

Table 9.1. Cross-corpus appearances of the $\dim(Z) = 2$ invariant.

Appearance	Quantity	Status
ZS-Q7 slow eigenvalue	$-2\mathbf{A}/\mathbf{Q} = -\dim(Z)\kappa^2$	PROVEN / DERIVED
ZS-U10 Schwinger coeff.	$C_\square = \dim(Z)/(4\pi) = 2/(4\pi)$	DERIVED
ZS-F19 KMS rapidity	$\frac{1}{2}\ln 2$, $\dim(Z) = 2$ channels	DERIVED
ZS-A12 Nelson–Kosterlitz	universal-jump integer 2	DERIVED

v2.1 executes the three-step programme — (1) derive the matching condition $C_0(\kappa^2)$ via controlled RG running from the ZS-S10 Planck-scale gauge bridge ($f^2 = \kappa^2 M_p^2$); (2) compute the resulting ANC and

$S_{34}(E)$; (3) compare to Zhang–Nollett–Phillips (2020) [4] — in §10. The result promotes $\delta \ln \gamma_b = -\kappa^2$ from HYPOTHESIS-strong to DERIVED-CONDITIONAL and makes the S_{34} prediction parameter-free.

§10. The $C_0 \leftrightarrow \kappa^2$ RG-Running Matching:

§10.1. Statement of the Closure and the Load-Bearing Issue

ZS-S10 §3.4 explicitly noted that “a complete RG analysis from M_Z to the Stückelberg scale is outside the scope of that paper.” The matching reduces to a single load-bearing question: does the geometric ratio $\kappa^2 = A/Q$ run between M_p and the nuclear scale? If it runs, the Planck-fixed value and the nuclear-scale ANC shift differ and the matching collapses. We prove it does not run (Theorem T10.6), so the matching is direct: the Planck-fixed κ^2 IS the nuclear-scale fractional ANC shift, with the only scale-dependent piece (the EFT contact-term reshuffling) being physically inert.

§10.2. Theorem T10.6 (κ^2 Non-Running, DERIVED-CONDITIONAL)

Statement. $\kappa^2 = A/Q$ acquires zero anomalous dimension under renormalization-group flow from M_p to the nuclear EFT scale; hence the Planck-scale matching value and the nuclear-scale value coincide.

Proof (two layers).

(i) Continuum Perturbative Protection (corpus, PROVEN-PERTURBATIVE). The Continuum Perturbative Protection Theorem (ZS-M13 §7A, equivalently ZS-M6 §7A) proves $L_{XY}^{\text{eff,direct}} = 0$ to all orders in perturbation theory, via a Ward–Takahashi identity assembled from four independently PROVEN inputs (ZS-M2 Lorentz algebra; ZS-F1/ZS-S1 action-level absence of X–Y coupling; ZS-Q5 frame invariance; ZS-F2 §4.2A Schur A_5 protection). The coefficient κ^2 multiplies the cross-sector (X–Y, Z-Spin-mediated) operator; because that channel is non-renormalized to all perturbative orders, κ^2 carries no perturbative anomalous dimension. The companion rank bound $\text{rank}(T_{XY}) \leq \dim(Z) = 2$ (ZS-Q7 Theorem 2, DERIVED) “survives the continuum limit because it depends only on $\dim(Z)$, not on the lattice spacing” (ZS-M17 §3.2) — the explicit scale-independence statement.

(ii) Appelquist–Carazzone decoupling (external, PROVEN [15]). The heavy Stückelberg mode of the ZS-S10 gauge bridge has mass $m_B \sim \kappa g_Y M_p$ (GUT-scale, ZS-S10 §3.3). By the Appelquist–Carazzone theorem [15], heavy-mass effects on light-field Green’s functions reduce to either (a) renormalization of light couplings or (b) operators suppressed by inverse powers of m_B . κ^2 enters as the dimensionless coefficient of the marginal protected cross-sector operator — channel (a), not power-suppressed — and by layer (i) is not renormalized. Hence κ^2 is RG-invariant from M_p to the nuclear scale. (Note: AC decoupling can be violated when a heavy field acquires a dimension-full coupling proportional to a symmetry-breaking VEV; here the cross-sector coupling κg_Y is dimensionless and the operator is protected, so the non-decoupling pathway is closed at the perturbative level.)

Status. DERIVED-CONDITIONAL on the perturbative scope of §7A (weak curvature $R \ll M_p^2$). The non-perturbative strong-curvature completion is the single residual OPEN (Gate S-T10.2), narrow and non-blocking: the nuclear scale (~ 71 MeV) is deep in the weak-curvature regime, so the numerical content is unaffected. **■**

§10.3. The EFT Matching Equation $C_0(\kappa^2)$

In ${}^3\text{He}\text{--}{}^4\text{He}$ cluster EFT at leading order (PDS scheme, Kaplan–Savage–Wise [16]) the short-range contact coupling runs as $C_0(\mu) = (4\pi/\mu_{\text{red}}) \cdot 1/(\mu - \gamma_b)$, where μ is the EFT renormalization scale and μ_{red} the reduced mass. The physical pole position γ_b is μ -independent: the μ -dependence of $C_0(\mu)$ cancels against the bubble-sum loop. The operator $T_{\{Y \rightarrow X\}}$ shifts the physical binding momentum, $\gamma_b \rightarrow \gamma_b(1 - \kappa^2)$. Propagating this RG-invariant shift through $C_0(\gamma_b)$ gives the matching condition

$$\begin{aligned}\delta \ln C_0(\mu) &= [\gamma_b/(\mu - \gamma_b)] \cdot (-\kappa^2) \quad [\text{scheme/scale-dependent; physically inert}], \\ \delta \ln \gamma_b &= -\kappa^2 \quad [\text{RG-invariant physical content; Theorem T10.6}].\end{aligned}$$

Because the physical observable (the ANC / pole residue) is scheme-independent — NLO halo-EFT capture cross-sections are confirmed regulator-independent (e.g. [5,17] and the regulator-independence results in the recent halo-EFT Coulomb-breakup literature) — the matching is well-defined at every μ : C_0 adjusts to reproduce exactly $\delta \ln \gamma_b = -\kappa^2$. Status: DERIVED (standard EFT + Theorem T10.6).

§10.4. From C_0 to S_{34} : Parameter-Free ANC Kinematics

The shallow-bound-state ANC obeys the Sparenberg–Capel–Baye relation [17] (compact algebraic equation connecting binding momentum, ANC, and effective-range expansion; equivalently derivable from the EFT causality integral identity): $C^2 = 2\gamma_b/(1 - \gamma_b r_0)$. Hence

$$\delta \ln C^2 = \delta \ln \gamma_b / (1 - \gamma_b r_0) = -\kappa^2 / (1 - \gamma_b r_0).$$

With $\gamma_b r_0 \approx 0.40\text{--}0.50$ (Zhang–Nollett–Phillips 2020 [4]) and $S_{34} \propto C^2$:

$$\delta \ln S_{34} = -0.73\% \text{ (LO, } r_0 \rightarrow 0) \text{ to } -1.5\% \text{ (with effective range).}$$

Every input is now LOCKED or external-measured: $\kappa^2 = A/Q$ (geometric, ZS-F2/ZS-M6), γ_b (AME2020 [12]), $\gamma_b r_0$ (Zhang et al. [4]). The S_{34} shift is therefore a parameter-free DERIVED-CONDITIONAL number, no longer merely an interesting falsifiable hypothesis. Status: DERIVED-CONDITIONAL (on Theorem T10.6 perturbative scope and the Zhang et al. effective-range input).

§10.5. Comparison to Zhang–Nollett–Phillips (2020) and External Consistency

The Zhang–Nollett–Phillips ANC determination carries a few-percent uncertainty; the predicted -0.73% to -1.5% shift lies within their error budget — consistent, not yet discriminating. The S_{34} prediction remains TESTABLE at the LUNA-MV frontier (S34-1). No conflict with external data arises: A’s prior derivation matches Planck 2018 ΛCDM and the H_0 ratio [13,14], and the ${}^8\text{B}/{}^9\text{C}$ existence (AME2020 [12]) is respected by bottleneck-selectivity (NC-T10.5).

§10.6. Version-Conflict (Dependency) Audit

The closure changes NO numerical value ($\delta \ln \gamma_b = -\kappa^2 = -0.728\%$ is identical to v1.0–v2.0). Cross-paper dependency trace:

Table 10.1. Version-conflict audit of the §10 closure (downstream-inert).

Upstream / downstream paper	Quantity	Effect of §10 closure
ZS-M1 (i-tetration fixed point)	$z^* = 0.43828 + 0.36059i$	Untouched (no numerical change)
ZS-Q7 (inter-sector generator)	$\lambda_{\text{slow}} = -2A/Q = -\dim(Z)\kappa^2$	Untouched; §10 uses it as PROVEN input
ZS-S10 (gauge bridge)	$f^2 = \kappa^2 M_p^2$	Completes ZS-S10 §3.4 deferred RG analysis

Upstream / downstream paper	Quantity	Effect of §10 closure
ZS-S1 / ZS-U1	$\alpha\Box = 11/93$; $r = 0.0089$	Untouched (ZS-S10 §3 backward-compat)
Gate Li-2 (nonlinear ${}^7\text{Li}$)	linear $+0.3\sigma$ / nonlinear -3.3σ	UNCHANGED — still FLAGGED (different gate)
${}^8\text{B}/{}^9\text{C}$ selectivity (NC-T10.5)	bottleneck-selective action	Untouched

§10.7. Anti-Numerology Pre-Registration for the Matching Step

Structural anti-numerology for the eigenvalue is inherited from §6 (Part A: 100% structural identity; Part C: 0.54% rate-law necessity). For the new matching step we pre-register: among random RG-invariant cross-sector ratios $r \in (0,1)$, the joint probability that r reproduces the Zhang–Nollett–Phillips ANC consistency AND equals $\mathbf{A}/\mathbf{Q}$ at the ZS-M6 $10^{-14}\%$ uniqueness window is $< 1\%$ (the ZS-M6 §2.2.2 uniqueness test already shows the three nearest candidates $\mathbf{A}/(\mathbf{Q}-\mathbf{Z})$, $3\mathbf{A}/\mathbf{Q}^2$, $\mathbf{A}$ are $226\times/765\times/8074\times$ worse). This MC is pre-registered; full execution is pending. Accordingly the matching-specific promotion carries an explicit anti-numerology-pending flag, while the eigenvalue itself (§6 Part A) is already structurally established at 100%.

§10.8. Epistemic Upgrade Summary

Item	v2.0 status	v2.1 status
$\delta\ln\gamma_b = -\kappa^2$	HYPOTHESIS-strong	DERIVED-CONDITIONAL
S_{34} shift	interesting falsifiable hypothesis	parameter-free DERIVED-CONDITIONAL (TESTABLE)
Gate S-T10.1 ($\text{C}_0 \leftrightarrow \kappa^2$)	OPEN	CLOSED-CONDITIONAL
Theorem T10.6 (κ^2 non-running)	— (not stated)	DERIVED-CONDITIONAL (new)
Residual OPEN	entire matching	Gate S-T10.2 (non-perturbative only; narrow)

Acknowledgements & Code Availability

PRyMordial (Burns, Tait, Valli [10]) was used under its public license for the BBN runs. All symbolic, Monte-Carlo, EFT, O-F19.6, ${}^8\text{B}/{}^9\text{C}$ -consistency, Bayes-factor, PRyMordial-gate, and (new in v2.1) §10 RG-matching / Theorem T10.6 scripts (35 checks, fixed random seeds) reproduce every figure herein and are available on request. No new free parameters were introduced; the §10 closure is parameter-free.

Appendix A. Numerical Inputs

All v2.0 rows are preserved verbatim (no-deletion rule); the final block adds the §10 closure inputs introduced in v2.1.

Quantity	Value	Source
$\mathbf{A} = \delta_X \cdot \delta_Y$	$(5/19)(7/23) = 35/437$	ZS-F2 (geometry)
$\kappa^2 = \mathbf{A}/\mathbf{Q}$; λ_{slow}	$35/4807$; $-70/4807$	ZS-M6; ZS-Q7
$\delta\ln\gamma_s = -\kappa^2$	-0.00728 (-0.73%)	This work §3

Quantity	Value	Source
$\gamma_3 r_0$ (^3He – ^4He eff. range)	0.40–0.50	Zhang et al. 2020 [4]
$\delta \ln S_{34}$ (predicted)	–0.7% to –1.5%	This work §4.2
^8B S_p ; B_{tot}/Q	0.137 MeV; 275	AME2020 [12]
^9C S_p ; B_{tot}/Q	1.299 MeV; 30.1	AME2020 [12]
Bayes factor $\ln \Lambda$	7.2 (very strong)	This work §6.5
PRyM baseline $^7\text{Li}/\text{H}$	5.49×10^{-10}	This work
H_0 ratio e^A (external check)	1.0834 vs Riess 73.04, Planck 67.36	[13,14]
$\gamma(\kappa^2)$ anomalous dim. (perturbative)	0 (exact)	This work §10.2; ZS-M13 §7A; [15]
$\delta \ln \gamma_b$ (RG-invariant)	$-\kappa^2 = -0.00728$	This work §10.3 (Theorem T10.6)
$\delta \ln C_0(\mu)$ (scheme-dependent)	$-\kappa^2 \cdot \gamma_b / (\mu - \gamma_b)$	This work §10.3; [16]
$C^2 = 2\gamma_b / (1 - \gamma_b r_0)$	ANC $\leftrightarrow$ binding momentum	Sparenberg et al. [17]
$\delta \ln S_{34}$ (parameter-free, v2.1)	–0.73% (LO) to –1.5% (eff. range)	This work §10.4

Appendix B. Review-Audit Record (v1.0 $\rightarrow$ v2.1)

All corrections adopted under the no-deletion rule. Items from v1.1, v1.2, v1.3, v2.0 preserved verbatim; v2.1 adds items 17–19 closing the $C_0 \leftrightarrow \kappa^2$ programme.

#	Issue found	Version	Resolution
1	Universal $B_{\square_0 \square} / Q$ ‘20–30’ FALSIFIED ($^8\text{B}=275$)	v1.1	§4.1 per-nucleus table
2	$\delta \ln \gamma_{\square} = -\kappa$ FACTUAL ERROR (correct: $-\kappa^2$)	v1.1	§3 $\delta \ln \gamma_{\square} = -\kappa^2$
3	$2.3\% \times 0\% \times 0.54\% = 0$ LOGIC ERROR	v1.1	§6.4 structural vs statistical; no product with zero
4	$-2A/Q$ convergence across U10, F19 OVERCLAIM (U10 = $\dim(Z)/4\pi$)	v1.1	§9 $\dim(Z) = 2$ convergence
5	S_{34} prediction magnitude $\sim 1\%$ not yet measured	v1.2	NC-T10.7; stated honestly
6	^9C ‘disappears’ (^9C is -44% , not full unbinding)	v1.2	§4.1 $^8\text{B}=-401\%$ unbinds; $^9\text{C}=-44\%$
7	$\ln \Lambda = 7.2$ ‘decisive’ ($\neq$ decisive, which requires >10)	v1.2	§6.5 stated as 7.2 (>5 , not >10)
8	^6Li reaction: $^6\text{Li}(p,\gamma) \rightarrow$ correct channel $^2\text{H}(\alpha,\gamma)^6\text{Li}$	v1.2	§4.2 corrected channel
9	S_{34} sensitivity ‘1–2’ uncited hand-wave	v1.3	§4.2 $S \propto \text{ANC}^2$ [6,7]; $1/(1-\gamma_{\square} r_0)$ basis
10	Selectivity theory only empirical (^8B), no theory	v1.3	§2.3 limiting-step $\rightarrow$ ^7Be bottleneck (with quantitative ^8B rate ratio)
11	$\ln \Lambda$ definition: H_0/H_1 undefined; product unjustified	v1.3	§6.5 explicit H_0/H_1 + independence caveat

#	Issue found	Version	Resolution
12	§5 O-F19.6 disconnected from main result	v1.3	§5 shared $p_{e\gamma}=(3,2,6)/11$ link sentence
13	H_0 external citation absent for A's BBN independence claim	v2.0	§1.1 table row 2: Riess 2022 [13], Planck 2018 [14] added; App. A last row
14	$\gamma\Box r_0 \approx 0.4\text{--}0.5$ uncited	v2.0	§3 'consistent with Zhang et al. 2020 [4]' added; App. A row added
15	$C_0 \leftrightarrow \kappa^2$ OPEN scope not clearly bounded	v2.0	§2.2 + NC-T10.8: RG running Planck $\rightarrow$ nuclear EFT explicitly stated as OPEN
16	Reference [7] duplicate (Costantini & Bemmerer both [7])	v1.3 $\rightarrow$ v2.0	References renumbered: Igamov [6], Costantini [7], Bemmerer [8], Adelberger [9], Burns [10], Fields [11], AME2020 [12]; Riess [13], Planck [14] added
17	$C_0 \leftrightarrow \kappa^2$ RG running	v2.0 $\rightarrow$ v2.1	§10 + Theorem T10.6: κ^2 non-running established via the Continuum Perturbative Protection Theorem (ZS-M13 §7A) + Appelquist–Carazzone decoupling [15]; Gate S-T10.1 $\rightarrow$ CLOSED-CONDITIONAL
18	$\delta \ln \gamma_b = -\kappa^2$ only HYPOTHESIS-strong; S_{34} shift not parameter-free	v2.1	§10.3–10.4: $\delta \ln \gamma_b = -\kappa^2 \rightarrow$ DERIVED-CONDITIONAL; matching scheme-dependence resolved [16]; via Sparenberg et al. [17] the S_{34} shift becomes parameter-free (-0.73% to -1.5%)
19	Residual non-perturbative protection left unbounded	v2.1	§10.2 + new Gate S-T10.2 (narrow, non-blocking): strong-curvature $R\text{--}M_P^2$ regime declared OPEN; nuclear scale ~ 71 MeV $\ll M_P$ confirms weak-curvature validity of the closure

References

- [1] T. Dent, S. Stern, C. Wetterich, Phys. Rev. D 76, 063513 (2007); arXiv:0705.0696.
- [2] V. De Vuyst, S. Eccles, P. A. Höhn, J. Kirklin, JHEP 07 (2025) 063; arXiv:2507.14131.
- [3] A. N. Gorban, O. Radulescu, A. Y. Zinovyev, Chem. Eng. Sci. 65, 2310 (2010); arXiv:0903.5072.
- [4] X. Zhang, K. M. Nollett, D. R. Phillips, J. Phys. G 47, 054002 (2020). [$^3\text{He}\text{--}^4\text{He}$ EFT, ANC, $\gamma\Box r_0$ values]
- [5] R. Higa, G. Rupak, A. Vaghani, Eur. Phys. J. A 54, 89 (2018). [halo-EFT leading order]
- [6] S. B. Igamov, R. Yarmukhamedov, Nucl. Phys. A 781, 247 (2007). [ANC method, $S_{34} \propto \text{ANC}^2$]
- [7] H. Costantini et al. (LUNA), Nucl. Phys. A 814, 144 (2008); arXiv:0809.5269. [LUNA S_{34} measurement]
- [8] D. Bemmerer et al. (LUNA), Phys. Rev. Lett. 97, 122502 (2006). [LUNA S_{34} ; $\sim 4\%$ precision]
- [9] E. G. Adelberger et al., Rev. Mod. Phys. 83, 195 (2011). [Solar fusion S-factors standard review; $^7\text{Be}(p,\gamma)^8\text{B}$ rate at BBN T]
- [10] A.-K. Burns, T. M. P. Tait, M. Valli, Eur. Phys. J. C 84, 86 (2024); arXiv:2307.07061. [PRyMordial]
- [11] B. D. Fields, Ann. Rev. Nucl. Part. Sci. 61, 47 (2011). [Lithium problem review]
- [12] M. Wang et al. (AME2020), Chin. Phys. C 45, 030003 (2021). [Nuclear binding energies, ^8B $S_\Box$, ^9C $S_\Box$]
- [13] A. G. Riess et al. (SH0ES), ApJ 934, L7 (2022). [$H_0^{\text{local}} = 73.04 \pm 1.04$ km/s/Mpc; external check for A's BBN independence]
- [14] Planck Collaboration, A&A 641, A6 (2020). [$H_0^{\text{cMB}} = 67.36 \pm 0.54$ km/s/Mpc; external check for A's BBN independence]
- [15] T. Appelquist, J. Carazzone, Phys. Rev. D 11, 2856 (1975). [decoupling theorem; basis for §10.2 κ^2 survival as a marginal protected cross-sector coefficient]
- [16] D. B. Kaplan, M. J. Savage, M. B. Wise, Nucl. Phys. B 534, 329 (1998); arXiv:nucl-th/9802075. [PDS scheme; scheme-dependent running of contact terms $C_0(\mu)$ vs scheme-independent pole/ANC content]

[17] J.-M. Sparenberg, P. Capel, D. Baye, Phys. Rev. C 81, 011601(R) (2010); arXiv:0907.5166. [ANC $\leftrightarrow$ binding momentum $\leftrightarrow$ effective-range relation $C^2 = 2\gamma_b/(1-\gamma_b r_0)$, used in §10.4]

Version History

v1.0 (Mar 2026): Operator $T_{\gamma \rightarrow x}$; PRyMordial gate with registered nonlinear ${}^7\text{Li}$ tension (Li-2).

v1.1 (Mar 2026): $\lambda \square^{\text{low}} = -\dim(Z)\kappa^2$; $\delta \ln \gamma \square = -\kappa^2$; per-nucleus P-T10.1; structural-vs-statistical §6.4; $\dim(Z) = 2$ convergence. Four review proposals corrected (App. B #1–4).

v1.2 (Mar 2026): First external-reviewer revision. Adds §1.1 A's chronological priority; §1.2 standard-physics translation; §2.2 ANC/Wilson coefficient restatement; §4.1 ${}^8\text{B}/{}^9\text{C}$ consistency test; §4.2 S_{34} prediction + LUNA-MV programme; §6.5 Bayes factor $\ln \Lambda = 7.2$;

v1.3 (Mar 2026): Second external-reviewer revision. Adds §2.3 limiting-step $\rightarrow {}^7\text{Be}$ theoretical basis; $\gamma \square r_0 = 0.40\text{--}0.50$ ANC sensitivity in §4.2; explicit H_0/H_1 + independence caveats in §6.5; shared- $p_{e\gamma}$ sentence in §5. Four items corrected (App. B #9–12).

v2.0 (Mar 2026): Third external-reviewer revision. Adds §1.1 H_0 external citations (Riess [13], Planck [14]); §3 / App. A $\gamma \square r_0 = 0.40\text{--}0.50$ citation (Zhang et al. [4]); §2.2 + NC-T10.8

v2.1 (Mar 2026): Adds §10 (Theorem T10.6: κ^2 non-running, DERIVED-CONDITIONAL) via the Continuum Perturbative Protection Theorem (ZS-M13 §7A) + Appelquist–Carazzone decoupling [15]; matching equation $\delta \ln \gamma_b = -\kappa^2$ (RG-invariant) with scheme-dependent $C_0(\mu)$ running [16]; parameter-free S_{34} shift via the Sparenberg–Capel–Baye ANC relation [17]; new epistemic tags DERIVED-PERTURBATIVE and CLOSED-CONDITIONAL; Gate S-T10.1 $\rightarrow$ CLOSED-CONDITIONAL; new narrow non-blocking Gate S-T10.2; NC-T10.8 superseded, NC-T10.9 / NC-T10.10 added; verification count $32 \rightarrow 35$. No v2.0 content deleted (additive only). Three items added (App. B #17–19).